Visualization # 1 (WFH Dashboard)

https://public.tableau.com/views/WFHDashboard_17813862670670/WFHDashboard?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Summary

For my dashboard I wanted to answer two questions at once: which states leaned hardest on working from home, and how common working from home actually was across the country. So I added two charts to the dashboard. A bar chart ranking the states and a histogram showing how all 3,220 counties are spread out.

Right at the top: South Dakota (about 10.4%), Montana, North Dakota, and Nebraska – the exact farm and ranch states that lit up on the map. The histogram tells the other half of the story. About 79% of all counties sat under 6% working from home, and only 1.6% ever hit 15% or more, with the typical county landing around 3.9%. Together, they make the point clear: in 2015, working from home was a rural thing, and even then it was rare. The Plains states aren't the norm, they're the outliers.

One important note on the bar chart: I averaged the counties evenly, which gives the little rural counties a loud voice. If I'd weighted by population instead, those Plains states would drop quite a bit (South Dakota falls from 10.4% to about 5.8%). I went with the plain average because it matches the rural story the rest of my project tells, but I wanted to flag the tradeoff.

Design

The dashboard combines two different worksheets on a single page. I used a horizontal bar chart for the state ranking because bars make it easy to compare one number across a list of categories, and a histogram for the distribution because that's the right way to show the shape of the data. They answer different questions, which is why I put them together.

I kept the same blue-teal sequential palette as my map so everything feels cohesive, and it stays colorblind-safe. I added a state filter so a viewer can focus on whatever states they want, and the bar chart's tooltip also shows each state's average commute time in minutes for a little extra context on hover.

Resources

p1-lessons-cheat-sheet-nd197-main.pdf
https://www.youtube.com/watch?v=9xqHA732LMA&feature=youtu.be

Visualization # 2 (WFH by County)

https://public.tableau.com/shared/S2JM4NH5S?:display_count=n&:origin=viz_share_link

Summary

The first thing I wanted to figure out was simply where in the country people actually worked from home back in 2015. So, I mapped every county in the U.S. and shaded each one by percent of its workers who work from home, the darker the county, the more remote workers.

What jumped out at me is how empty most of the map is. The average county sat at only about 4.6%, so for most of America, working from home in 2015 wasn't really a thing. And the places that did light up

weren't the big tech cities I assumed they'd be. They were rural counties out West and across the Great Plains. Once I thought about it, that made sense. A farmer or rancher technically "works from home" because home is the farm. What makes this interesting is that just a few years later, that number would shoot up for a totally different reason (the pandemic) and a totally different kind of worker (office workers).

Design

I chose a filled map because the data ties to real places, every row is a county, so a map lets you see the pattern in a couple of seconds instead of scrolling through a list of 3,000 plus county names. The rural vs. city split basically reads itself.

For color, I used a single-color palette (blue into teal), light for low percentages and dark for high. I picked a sequential scale on purpose instead of a diverging one: the numbers only run from 0% up, they never go negative, so there's no middle point to diverge from. Light to dark is the honest way to show less to more. I also stayed away from any red and green colors because that is the toughest combo for colorblind readers, and blues and teals stay readable across the common types of color blindness. Also, most counties bunch up at the low end, so I capped the color scale at 15% to stretch out the contrast and make the rural hotspots actually pop instead of washing out.

Resources

p1-lessons-cheat-sheet-nd197-main.pdf
https://www.youtube.com/watch?v=9xqHA732LMA&feature=youtu.be

Visualization # 3 (WFH vs Income)

https://public.tableau.com/views/WFHvsIncome/WFHvsIncome?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Summary

Next, I wanted to test something I assumed was true: that working from home is a wealthy person's thing. So, I plotted every county with median household income along the bottom and the percent working from home up the side. One dot per county, plus a trend line.

Turns out my assumption was incorrect. The dots sit in one big low cloud, and the trend line barely tilts up. The R-squared was only about 0.04, meaning income explains just ~4% of why some counties work from home more than others. The poorest quarter of counties averaged 3.5% working from home and the richest quarter only 5.3%, so money nudges it a little but doesn't drive it. Back in 2015, your paycheck basically wasn't what decided whether you worked from home.

Design

I picked a scatter plot because I'm comparing two number columns and want to see if they move together. The trend line makes the lack of relationship easy to read. I kept every dot a single color on purpose; adding more colors wouldn't mean anything here and would just create clutter. I made the dots see-through with a thin border so the dense middle doesn't turn into a solid blob, and I labeled both axes with their units (dollars and percent) so it's clear what you're looking at.

Resources

p1-lessons-cheat-sheet-nd197-main.pdf

https://www.youtube.com/watch?v=9xqHA732LMA&feature=youtu.be